

SMART WASTE MANAGEMENT IN GOA USING AI



AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN ASSAGAO BARDEZ GOA

TEAM NAME:- AI LEARNERS

MEMBERS NAME:-

SALONI UDAY MALIK

MAYUR RAMAKANT GHADI

BRANCH :- COMPUTER



Problem Statement

Goa faces challenges like overflowing garbage bins, poor segregation, illegal dumping, and overburdened waste treatment plants. These issues affect cleanliness, worker safety, and the environment. Current systems lack real-time monitoring and public engagement. This project aims to solve these issues using AI technologies to create a smarter and cleaner Goa



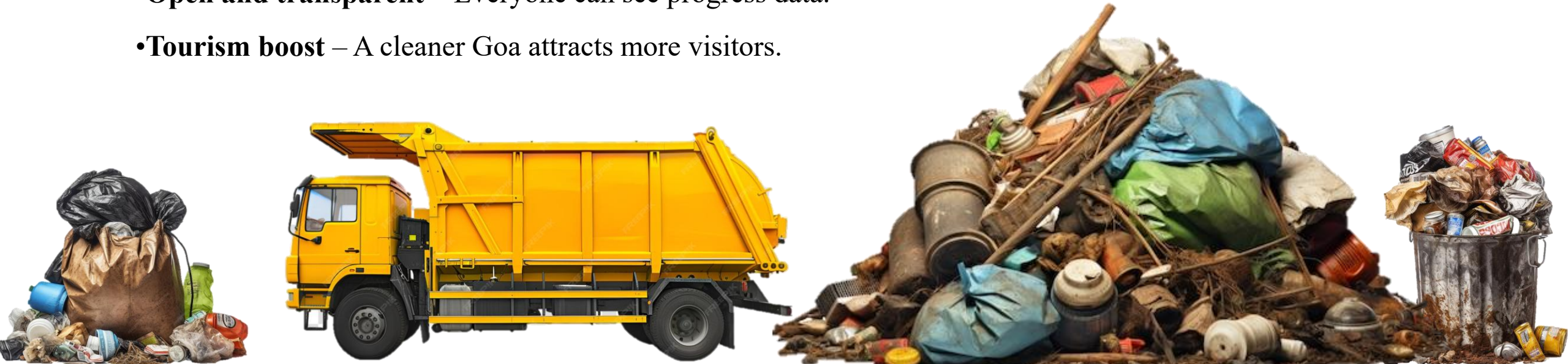
Solution Overview

- **Catching Illegal Dumping:** AI uses CCTVs to detect dumping, capture details, and alert municipal teams instantly.
- **Smarter Garbage Collection:** Citizens view pickup schedules, request extra pickups, and track complaints live.
- **Real-Time Truck Tracking:** Live maps show truck locations, ETAs, and missed stops for full coverage.
- **Rewards & Leaderboard:** Points, prizes, and leaderboards make cleanliness a fun community challenge.
- **Cleanup Event Hub:** Post events, recruit volunteers, and share before–after results for points.
- **Education for Change:** Fun, multilingual videos and quizzes teach waste segregation and eco-habits.
- **Complaint Box:** Report issues in seconds with photos and locations, and track resolution live.
- **Waste Data Transparency:** Open real-time stats on collection, segregation, and recycling build trust.



Expected Outcome / Impact

- **Stops illegal dumping** – Catches offenders fast using CCTV alerts.
- **Better garbage collection** – Request pickups and track them live.
- **Motivates good habits** – Points, prizes, and leaderboards.
- **Teaches in local languages** – Videos, guides, and quizzes in Konkani, Marathi, Hindi, and English.
- **Cleaner, healthier Goa** – Less waste, fewer diseases, better environment.
- **Community power** – People join cleanups and report issues.
- **Open and transparent** – Everyone can see progress data.
- **Tourism boost** – A cleaner Goa attracts more visitors.



Target Audience



- **Local Residents** – Individuals and families who generate household waste and want better collection services.
- **Municipal Workers & Authorities** – For monitoring garbage trucks, complaints, and illegal dumping.
- **Event Organizers & NGOs** – Groups that run cleanup drives or awareness programs.
- **Students & Youth** – To participate in challenges, spread awareness, and volunteer in events.
- **Businesses & Shops** – Especially those producing commercial waste and needing timely pickups.
- **Tourists & Visitors** – To know proper waste disposal rules and report issues.



Thank You!

Project: AI-Powered Smart Waste Management for Goa

Institution: *Agnel Institute of Technology and Design Assagao Goa*

Presented by:

Saloni uday malik

Mayur Ramakant Ghadi



“Building cleaner, smarter cities with the power of AI.”